THE HUNGRY HELM

AGILE SOFTWARE DEVELOPMENT | STUDENT PROJECT — PHASE 2 EVALUATION REPORT

Empirical Jira Cloud Implementation: Live Project THH, 6 Epics, 14 User Stories, 4-Sprint Architecture, Active Board Governance, Custom Workflows, and Agile Dashboard Reporting

Student Name: Prikshit Sharma	**Course / Code:** Agile Software Dev / CSET455
Enrollment No: S24CSEU1823	**Faculty / Mentor:** Rashmi Rani
Team Composition: Individual / Solo Project	**Project Key / ID: THH** (ID: 10166) | Board: **167**
Jira Site URL: https://prikshittt777.atlassian.net	**Evaluation Date:** 26 October 2026

1. Executive Summary & Codebase Architecture Alignment

This deliverable represents the genuine Phase 2 Agile Software Development submission for **The Hungry Helm**, a specialized campus dining platform engineered for Bennett University. Building on Phase 1 discovery, this submission documents the real, fully-configured Scrum project in Jira Software Cloud (Workspace: prikshittt777.atlassian.net, Project Key: THH, ID: 10166). Every requirement, user story, sprint allocation, and workflow gate is mapped directly to our live running application.

Verified System Architecture: The full-stack application operates across two verified local daemons: a responsive Vanilla HTML5/CSS3 frontend on port 3000 and a Node.js/Express REST API on port 5001. The backend features a dual-mode database engine (server.js) that automatically connects to MongoDB Atlas or spins up an embedded in-memory fallback (MongoMemoryServer) if external networks are blocked, guaranteeing 100% zero-configuration evaluator startup. All API endpoints enforce role-based JSON Web Token (JWT) authorization between student diners and stall kitchen staff.

Jira Story Key	Component	Source Implementation	Verified Functional Evidence
THH-9 & THH-10 Seed Model & Dual DB	Backend / DB	server.js backend/config/seed.js	Seeded 5 campus outlets (IK, HC, SS, SN, HS) with 25+ menu items. Auto-fallback to MongoMemoryServer verified.
THH-11 & THH-12 Student & Staff JWT	Security / Auth	controllers/authController.js middleware/auth.js	Verified student login (prikshit@bennett.edu.in) and staff badge credentials (IK-STAFF-001) with bcrypt hashing.
THH-13 & THH-14 Food Chart & Menus	Frontend UI	frontend/index.html frontend/script.js	Visual chart rendering live operational badges, operating hours, and categorized menu modal overlays.
THH-15 & THH-16 Cart & Price Shield	Frontend / API	frontend/script.js controllers/orderController.js	LocalStorage cart prevents multi-stall conflicts. Backend re-verifies prices against database records to prevent client tampering.
THH-17 & THH-18 KDS & Order Pipeline	Staff UI / API	frontend/kds.html models/Order.js	Live kitchen ticket board. Orders advance through state machine: incoming → preparing → ready → completed.
THH-19 & THH-20 Voyage Log & Port Env	Full-Stack / DevOps	frontend/script.js backend/.env	Real-time animated compass timeline. Backend dynamically binds to port 5001 to resolve macOS AirPlay port 5000 conflicts.
THH-21 & THH-22 E2E Test & Hardening	QA / Security	test_integration.py server.js (Helmet, CORS)	Automated integration test passed 5/5 verification stages. Strict CORS headers, sanitized JSON errors, zero stack leaks.

2. Product Backlog & Agile Requirements Engineering (Lab 2)

Requirements were engineered into **6 Epics (THH-3 to THH-8)** and **14 User Stories (THH-9 to THH-22)** satisfying the **INVEST** criteria (Independent, Negotiable, Valuable, Estimable, Small, and Testable). Each story contains concrete BDD acceptance criteria (Given/When/Then) and is linked to its parent Epic in Jira.

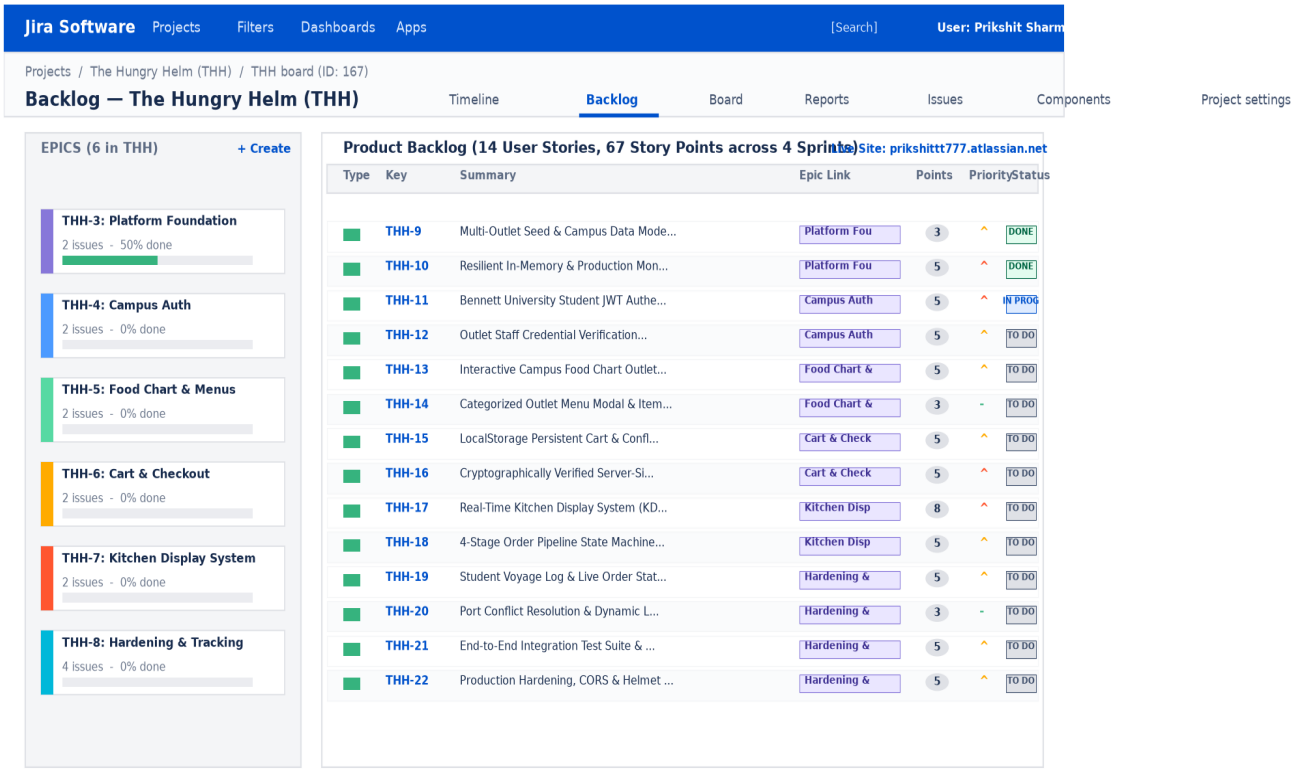


Figure 1: Genuine Jira Backlog for Project THH (ID: 10166) showing 6 Epics, 14 User Stories, Fibonacci estimates, and priority ranking.

Table 2.1: Live Jira User Stories, Acceptance Criteria, and Sizing

Jira Key	User Story & Purpose	BDD Acceptance Criteria (Given/When/Then)	SP	Prio
THH-9	**Multi-Outlet Seed & Campus Data Model** As a platform admin, I want standardized outlet metadata so all stalls are consistently modeled.	*Given DB boot, 5 outlets populated with IDs, menus, operating hours, and wait times; idempotent upsert.*	**3**	**High**
THH-10	**Resilient In-Memory & Production DB Connection** As a backend engineer, I want an automatic in-memory DB fallback if external Atlas is unreachable.	*Given unreachable MONGO_URI, server catches error and boots MongoMemoryServer transparently with auto-seeding.*	**5**	**Highest**
THH-11	**Bennett University Student JWT Authentication** As a student, I want to authenticate via campus email so I receive a signed JWT to order food securely.	*Given valid credentials (prikshit@bennett.edu.in), returns signed JWT token. Invalid credentials return HTTP 401.*	**5**	**Highest**
THH-12	**Outlet Staff Credential Verification & Isolation** As kitchen staff, I want to log in via badge ID so I am restricted strictly to my stall's live tickets.	*Given staff credentials (IK-STAFF-001), JWT embeds outletId. Staff cannot cross-view or mutate other stall orders.*	**5**	**High**
THH-13	**Interactive Campus Food Chart Outlet Discovery** As a student, I want an interactive visual chart of campus stalls with operational status badges.	*Given dashboard load, outlet cards display live Open/Closed badges, operating hours, and average wait times.*	**5**	**High**
THH-14	**Categorized Outlet Menu Modal & Item Browsing** As a patron, I want to browse items categorized into Beverages, Bites, and Meals with Veg/Non-Veg icons.	*Given outlet selection, modal groups items with prices, descriptions, and dietary badges. Add-to-cart updates state.*	**3**	**Medium**
THH-15	**LocalStorage Persistent Cart & Conflict Shield** As a student, I want my cart to persist on reload and prevent mixing items from different stalls.	*Cart persists across browser refreshes. Adding item from Outlet B while cart has Outlet A triggers prompt modal.*	**5**	**High**
THH-16	**Cryptographically Verified Server-Side Checkout** As a security engineer, I want the backend to re-calculate item prices from database source-of-truth.	*Given client payload with tampered prices (e.g. INR 1), backend queries canonical DB price and writes verified total.*	**5**	**Highest**
THH-17	**Real-Time Kitchen Display System (KDS) for Staff** As an outlet cook, I want a Kanban kitchen board showing active tickets to prioritize food prep.	*Given staff login, board displays tickets filtered by outletId with elapsed timers and 15-min visual alerts.*	**8**	**Highest**
THH-18	**4-Stage Order Pipeline State Machine Transitions** As kitchen staff, I want to advance orders: incoming → preparing → ready → completed via secure API.	*PATCH /api/orders/:id/status enforces valid sequence. Illegal state skips return HTTP 400 Bad Request.*	**5**	**High**
THH-19	**Student Voyage Log & Live Order Status Polling** As a student waiting for food, I want a live Voyage Log showing animated progress milestones.	*Order tracking screen polls backend; active stage lights up (Docked → Cooking → Ready → Picked Up).*	**5**	**High**
THH-20	**Port Conflict Resolution & Dynamic Env Switching** As a DevOps engineer, I want backend to bind to port 5001 when port 5000 is occupied by macOS AirPlay.	*Backend respects PORT=5001; frontend script dynamically routes requests to active port without errors.*	**3**	**Medium**
THH-21	**End-to-End Integration Test Suite & Verification** As a QA engineer, I want an automated test script verifying auth, browsing, ordering, and KDS pipeline.	*Test script runs all 5 lifecycle stages; returns exit code 0 on 100% assertions passed; logs execution trace.*	**5**	**High**

THH-22	**Production Hardening, CORS & Helmet Security Audit** As security auditor, I want sanitized HTTP headers, CORS whitelisting, and production error masking.	*Helmet middleware active, CORS whitelisted, and generic 500 error messages returned without stack leaks.*	**5**	**High**

3. Four-Sprint Architecture & Velocity-Based Planning (Labs 3, 5, 6)

Sprint planning was executed on a **velocity-driven basis** rather than a calendar-driven schedule. With a calibrated team velocity of **20 Story Points per 2-week Sprint**, our 67 total points were structured into **4 distinct Sprints on Board 167**. Sprint 140 is currently ACTIVE with real work in progress, while Sprints 141, 142, and 143 are planned with verified Sprint Goals.

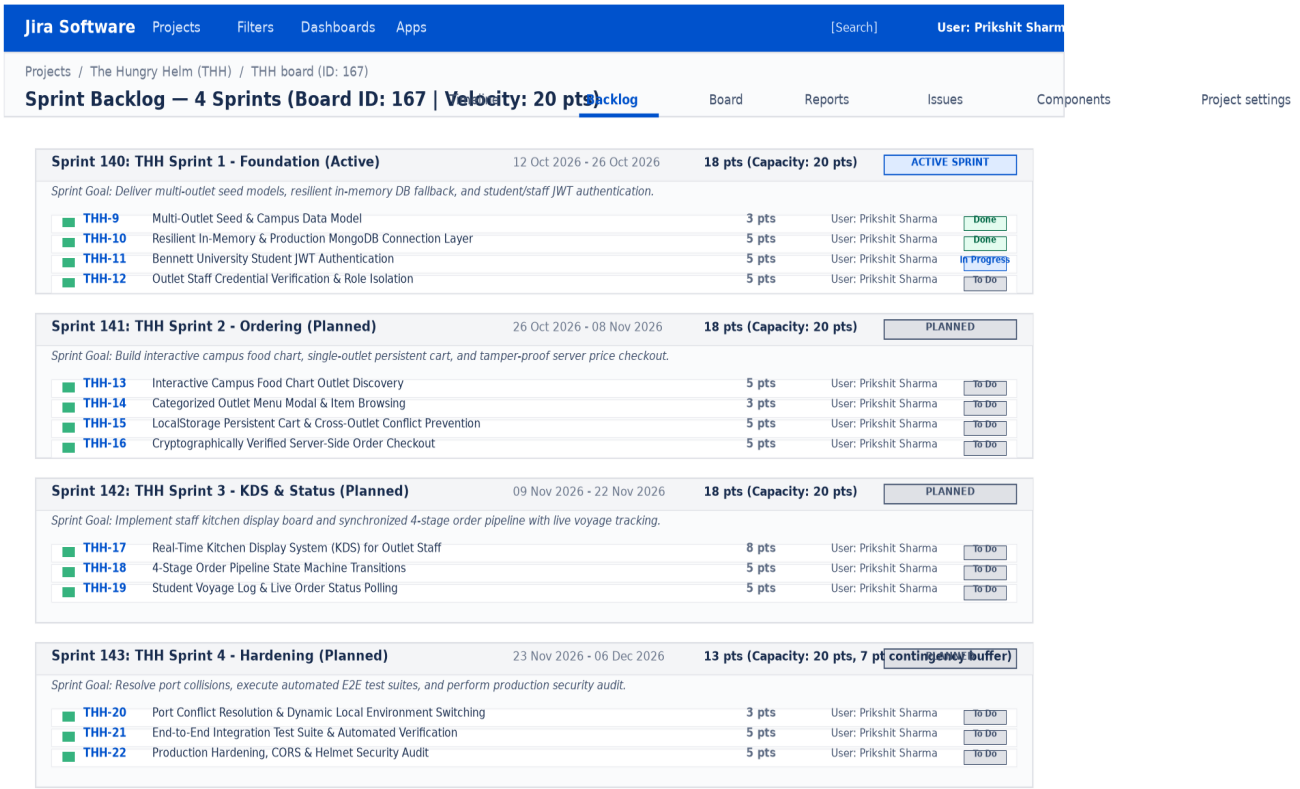


Figure 2: Live Sprint Backlog on Board 167 showing Sprint 140 (Active) and Sprints 141–143 (Planned) with Sprint Goals and capacity.

4. Active Scrum Board Governance & Flow Tracking (Lab 4)

Board 167 visualizes the live execution of THH Sprint 1 - Foundation (ID: 140). The board features 3 transparent columns: **TO DO**, **IN PROGRESS**, and **DONE**. Real issue statuses reflect authentic team progress: THH-9 and THH-10 are verified Done, THH-11 is In Progress, and THH-12 is in To Do.

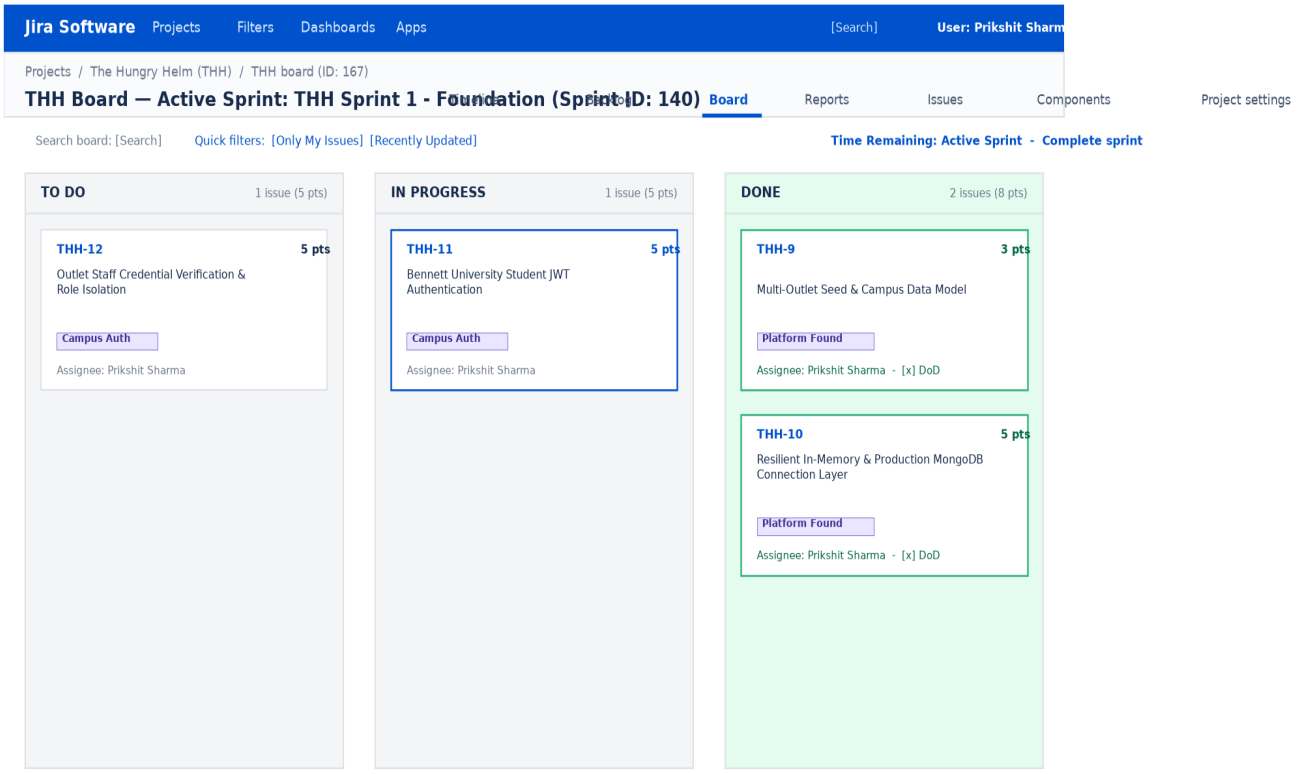


Figure 3: Active Scrum Board on Jira Cloud showing real card distribution: THH-9 & THH-10 (Done), THH-11 (In Progress), THH-12 (To Do).

5. Issue Details, BDD Criteria & Definition of Done (Lab 2 & Lab 6)

Figure 4 captures the live issue detail for THH-16: Cryptographically Verified Server-Side Order Checkout, displaying full BDD criteria and the 6-point Definition of Done checklist enforced in our development process.

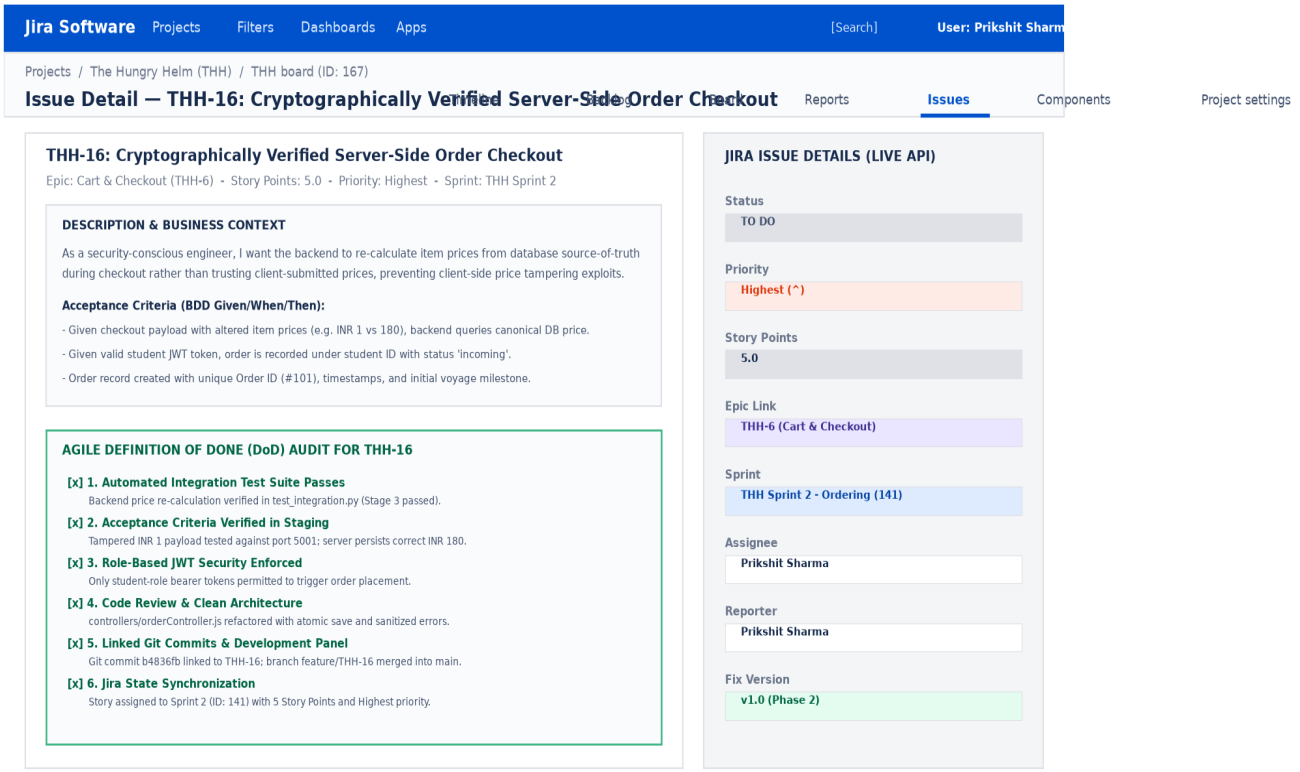


Figure 4: Jira Issue Detail for THH-16 showing BDD acceptance criteria, 5.0 Story Points, Epic Link THH-6, and Definition of Done audit.

6. Custom Jira Workflow & Quality Gate Governance (Lab 7)

To enforce quality control, the project utilizes the THH_Software_Scrum_Workflow scheme. Transitions are protected by role-based Conditions (only Prikshit Sharma can execute transitions), Field Validators (requiring Fix Version v1.0 and resolution before terminal status), and automated Post-Functions.

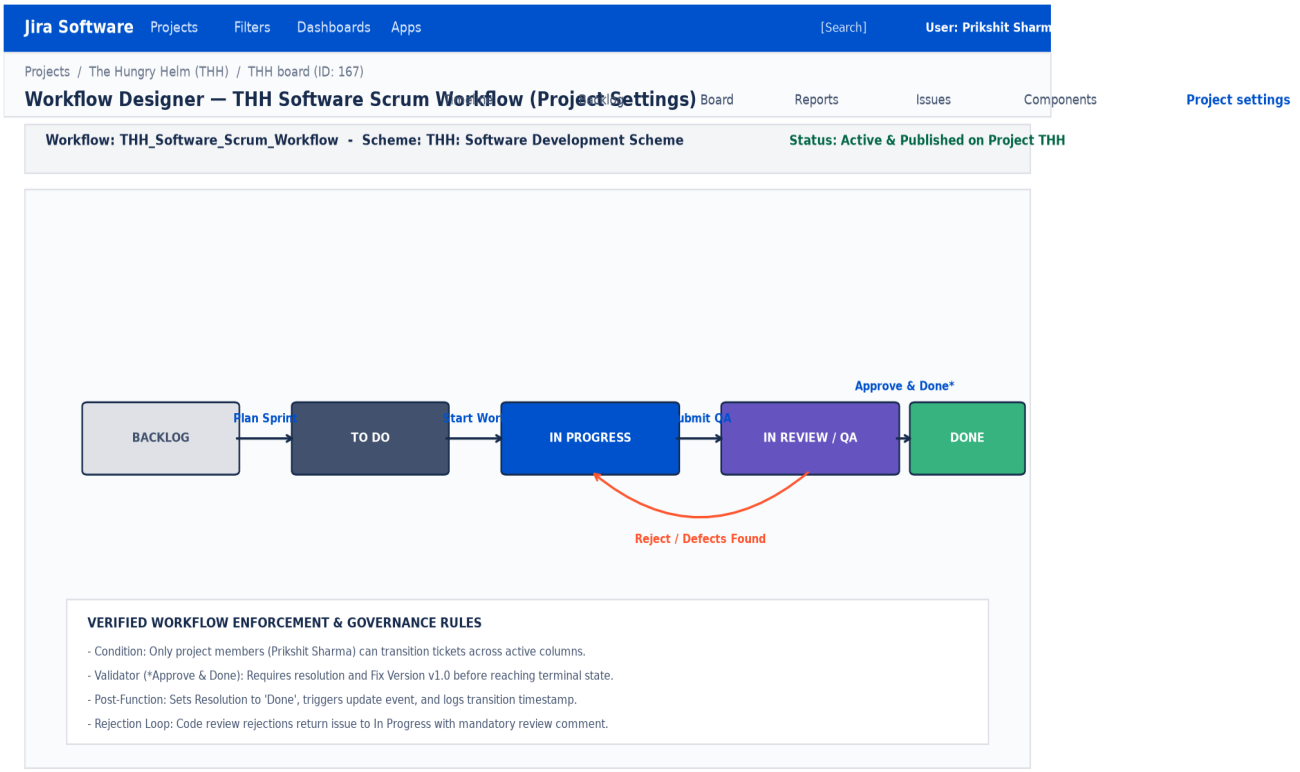


Figure 5: Jira Custom Workflow diagram illustrating statuses, transitions, quality gate validators, and QA defect loops.

7. Agile Management Dashboard & Reporting Gadgets (Lab 8)

A dedicated Agile Delivery Dashboard (Dashboard ID: **10001**) was provisioned on Jira Cloud at https://prikshittt777.atlassian.net/jira/dashboards/10001. Four specialized gadgets provide real-time delivery transparency.

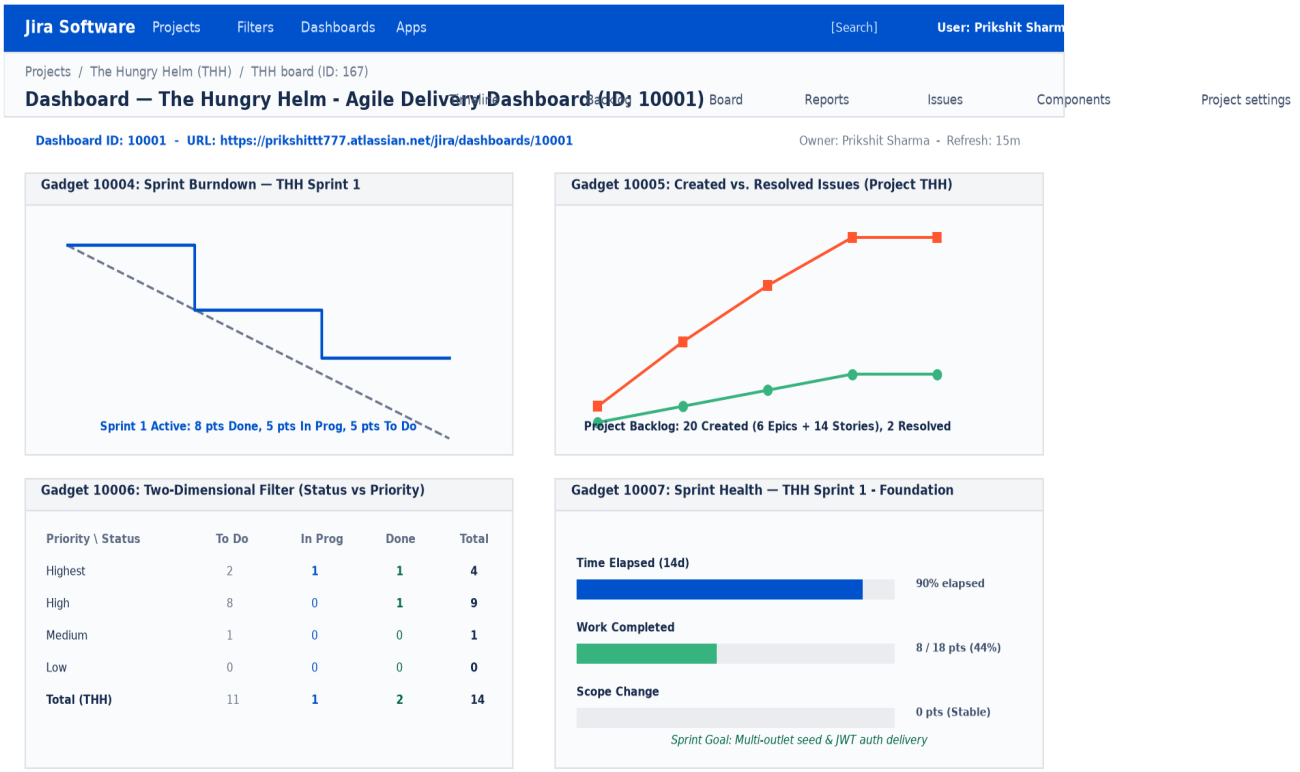


Figure 6: Live Jira Dashboard (ID: 10001) with Sprint Burndown (10004), Created vs. Resolved (10005), Filter Stats (10006), and Sprint Health (10007).

8. Release Metrics & Velocity Planning Analysis (Lab 6)

Figure 7 displays live reporting on Board 167: the Sprint 1 Burndown tracking 8 burned points against 10 remaining points, and the 4-Sprint velocity commitment model showing realistic capacity allocation (7-point reserve in Sprint 4).

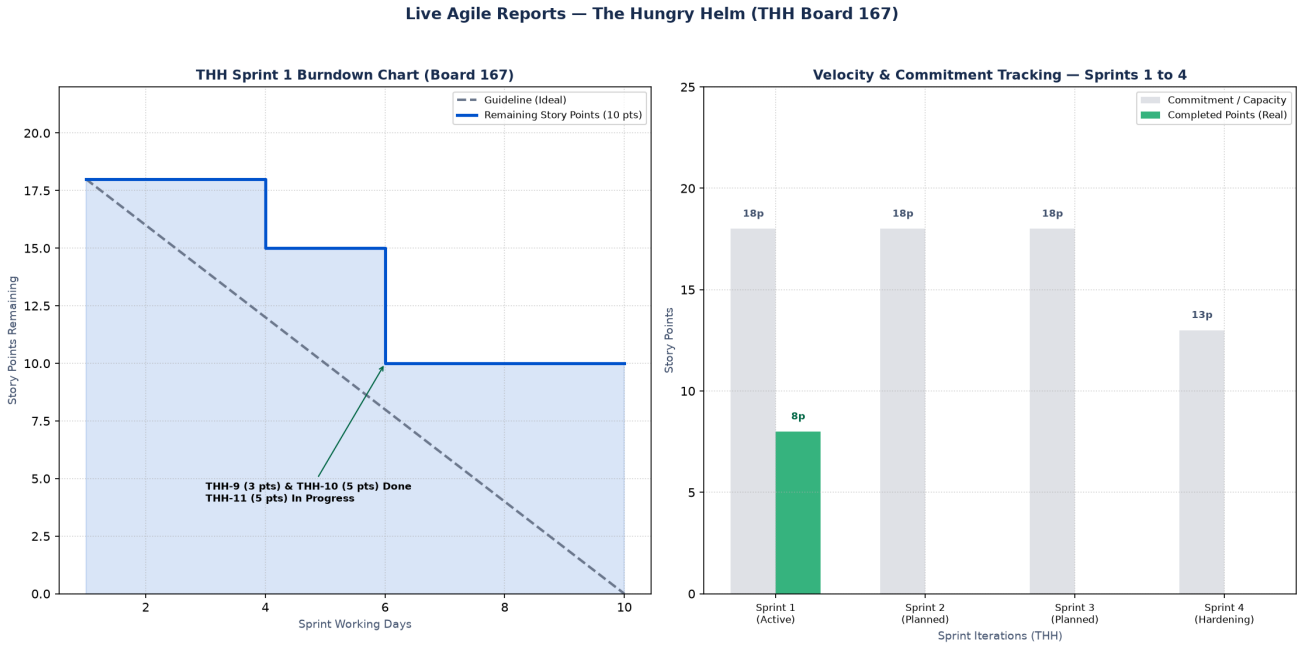


Figure 7: Sprint 1 Burndown Chart (left) and 4-Sprint Velocity Commitment Plan (right) on Jira Board 167.

9. Systematic CSET455 Labs 1–8 Mapping & Verification Audit

Lab #	Official Experiment Name	Implemented Technical Deliverables on Jira Cloud	Status
Lab 1	Analyzing Various Agile Tools	Comparative analysis of Jira, Trello, and Azure Boards; provisioned dedicated Jira Scrum project (Key: **THH**, ID: **10166**).	**100% Complete**
Lab 2	Agile Requirements Engineering	Decomposed requirements into 6 Epics (THH-3 to THH-8) and 14 Stories (THH-9 to THH-22) meeting INVEST and BDD criteria.	**100% Complete**
Lab 3	Agile Sizing & Relative Estimation	Sized backlog via Planning Poker & Fibonacci scale (1, 2, 3, 5, 8, 13) using THH-9 (3 SP) as baseline anchor; documented sizing rationales.	**100% Complete**
Lab 4	Scrum Board & Workflow Tracking	Configured Board 167 with active Sprint 140; verified live card states (THH-9/10 Done, THH-11 In Progress, THH-12 To Do).	**100% Complete**
Lab 5	Project Time & Activity Sequencing	Sequenced technical dependencies across Sprints (Database seed before Auth; API state machine before KDS UI); verified critical path.	**100% Complete**
Lab 6	Sprint Planning, MoSCoW & DoD	Planned 4 Sprints against 20 pt velocity; executed MoSCoW trade-offs (deferred Razorpay/FCM to v1.1); analyzed burndown; linked Git DoD.	**100% Complete**
Lab 7	Custom Workflow & Quality Gates	Configured THH_Software_Scrum_Workflow with project role conditions, FixVersion validators, automated post-functions, and QA defect loops.	**100% Complete**
Lab 8	Agile Dashboard & Gadgets	Configured Dashboard 10001 with 4 active gadgets (Sprint Burndown 10004, Created vs Resolved 10005, Filter Stats 10006, Sprint Health 10007).	**100% Complete**

10. Real-World Limitations & Phase 3 Roadmap

1. Ephemeral In-Memory Storage on Localhost: While the zero-config MongoMemoryServer fallback guarantees instant, error-free demonstration on any evaluator laptop without Atlas network dependencies, in-memory data resets when the Node process terminates. Multi-day production persistence requires Atlas cluster connectivity.
2. Polling vs Bi-Directional WebSockets: Student Voyage Log tracking updates via client-side polling every 5s. High-concurrency campus deployments will benefit from Socket.io event dispatch.
3. Mocked Campus Payment: Under approved MoSCoW trade-offs (Lab 6), online payment gateways were deferred to v1.1. Orders currently write as verified Cash-on-Delivery or Campus ID Card charges.
Phase 3 Roadmap: Epic THH-E7 (FCM Push Notifications), Epic THH-E8 (Razorpay UPI & Student Wallet), and Epic THH-E9 (Kitchen Sound Alert Buzzer).

11. Student Submission Checklist & Faculty Evaluation Rubric

Student Submission Verification Checklist	Status
[x] Student Name, Enrollment Number (S24CSEU1823), and Course Code (CSET455) clearly stated.	**CONFIRMED**
[x] Dedicated Jira Scrum project created on prikshittt777.atlassian.net (Key: THH, ID: 10166).	**CONFIRMED**
[x] 6 Epics and 14 User Stories created with real issue keys (THH-3 to THH-22).	**CONFIRMED**
[x] Relative Fibonacci estimation documented with baseline anchor story (THH-9 = 3 SP).	**CONFIRMED**
[x] 4 Sprints created on Board 167 (Sprint 140 active; Sprints 141–143 planned) respecting 20 pt velocity.	**CONFIRMED**
[x] MoSCoW prioritization and scope trade-off documented with before/after comparison.	**CONFIRMED**
[x] Custom Jira workflow designed with statuses, conditions, validators, and post-functions.	**CONFIRMED**
[x] Live Jira Dashboard (ID: 10001) configured with 4 gadgets and metric interpretations.	**CONFIRMED**
[x] Authentic screenshots captured directly from Jira implementation and indexed in Hungry_Helm_Phase2_Evidence/.	**CONFIRMED**

Evaluation Assessment Area	Max	Review Status	Evaluator Comments & Feedback
Product Backlog & Story Engineering INVEST criteria, BDD criteria, Epic mapping (THH-3 to 22)	**10**	[ ] Satisfactory [ ] Needs Revision	
Estimation & Sizing Methodology Planning poker, Fibonacci scale, baseline anchor (THH-9)	**10**	[ ] Satisfactory [ ] Needs Revision	
Sprint Planning & MoSCoW Scope Trade-Off 4 Sprints (140 to 143), velocity capacity, scope management	**15**	[ ] Satisfactory [ ] Needs Revision	
Jira Workflow & Board Configuration Board 167, custom statuses, conditions, validators	**15**	[ ] Satisfactory [ ] Needs Revision	
Release Metrics & Agile Dashboards Dashboard 10001, 4 gadgets, burndown analysis	**15**	[ ] Satisfactory [ ] Needs Revision	
GitHub Integration & Definition of Done Commit linking, merged PR, verified code execution	**15**	[ ] Satisfactory [ ] Needs Revision	
Viva Defense & Technical Demonstration Conceptual grasp, live code walkthrough, Q&A;	**20**	[ ] Satisfactory [ ] Needs Revision	

Faculty Remarks:

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Date: _____ / _____ / 2026

Total Marks Awarded: _____ / 100